

Lecture 11

The Lieb–Liniger gas

Objectives. Course map: G3, step 1 (the model). The method of [Lecture 10](#) works for particles on a line. Bosons with a contact interaction collide pairwise, each collision multiplies the wave by a phase, and the Bethe equations follow as before — this time with a complete proof that the wavefunction is an eigenfunction. For repulsive contact every root is real, and every set of quantum numbers gives exactly one state. This is the gas of the quantum Newton’s cradle, and the model whose thermodynamics [Lecture 16](#) and [Lecture 17](#) compute.

11.1 The question

[Lecture 10](#) solved a lattice model. [Lecture 0](#) asked why a one-dimensional gas might not thermalise, and pointed at the gas of the cradle. So the question is

does the Bethe ansatz solve an interacting gas — and why only in one dimension?

Yes, for bosons with a repulsive contact interaction ([Theorem 11.3](#)), with real roots ([Proposition 11.4](#)) and one state per set of quantum numbers ([Proposition 11.5](#)); and only in one dimension, because only there do two-body collisions keep the set of momenta (§ 11.5).

11.2 The gas

N identical bosons move on a ring of length L and interact only when they touch:

$$H = - \sum_{j=1}^N \frac{\partial^2}{\partial x_j^2} + 2c \sum_{i<j} \delta(x_i - x_j), \quad c > 0. \quad (11.1)$$

This is the *Lieb–Liniger* model (1963), repulsive for $c > 0$, in units $\hbar = 2m = 1$: a free particle of momentum k has energy k^2 and velocity $2k$.

It is the gas § 0.2 promised. In the thin tubes of the quantum Newton’s cradle ([Figure 2](#)) the transverse motion is frozen, and the atoms’ short-range interaction becomes, along the tube, a delta of a strength fixed by the scattering length and the tube width (Olshanii 1998) · *cited*. Between collisions the particles are free; the delta imposes a condition wherever two of them meet.

11.3 Two bosons

The contact condition

For two particles use the centre of mass $X = \frac{1}{2}(x_1 + x_2)$ and the separation $r = x_2 - x_1$. Then

$$-\partial_1^2 - \partial_2^2 = -\frac{1}{2}\partial_X^2 - 2\partial_r^2, \quad H = -\frac{1}{2}\partial_X^2 - 2\partial_r^2 + 2c \delta(r). \quad (11.2)$$

Integrate $H\psi = E\psi$ over $-\epsilon < r < \epsilon$ and let $\epsilon \rightarrow 0$: $-2[\partial_r \psi]_{r=0^-}^{r=0^+} + 2c\psi|_{r=0} = 0$. (ψ itself must be continuous: a jump would put a δ' into $\partial_r^2 \psi$ that nothing cancels.) So the r -derivative jumps by $c\psi$. For bosons ψ is even in r , its derivative odd, and the jump is twice the one-sided value. With $\partial_r = \frac{1}{2}(\partial_2 - \partial_1)$ this is the *contact condition*

$$(\partial_2 - \partial_1 - c)\psi = 0 \quad \text{at } x_2 = x_1, \text{ approached from } x_1 < x_2. \quad (11.3)$$

A symmetric ψ is fixed by its values on $x_1 < x_2$, so the problem is: *solve the free equation on $x_1 < x_2$, subject to (11.3) on its edge* (Figure 11.1).

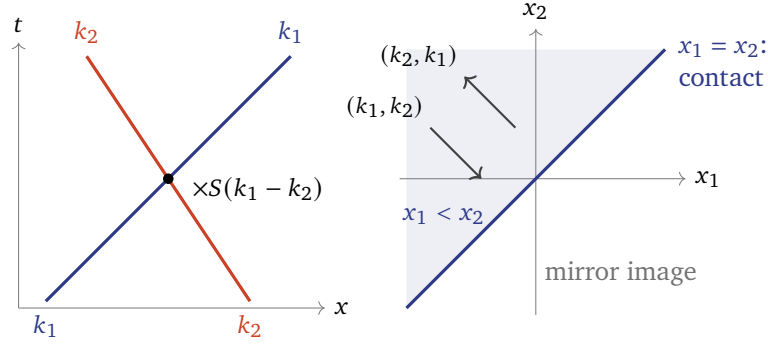


Figure 11.1: Two bosons. Left: the two momentum lines in space and time; the faster comes from the left, and after the collision the same two momenta travel on. All the collision can do is multiply the wave by $S(k_1 - k_2)$. Right: the configuration plane. The wavefunction is free on $x_1 < x_2$; on its edge the delta imposes (11.3), tying the wave (k_2, k_1) to the incoming (k_1, k_2) . The other half is the mirror image.

The two-body S-matrix

On $x_1 < x_2$ try, as in [Lecture 10](#),

$$\psi(x_1, x_2) = A_{12} e^{i(k_1 x_1 + k_2 x_2)} + A_{21} e^{i(k_2 x_1 + k_1 x_2)}, \quad E = k_1^2 + k_2^2. \quad (11.4)$$

Both waves solve the free equation with the same energy; only the contact condition is left.

Proposition 11.1 (The two-body S-matrix). *The wavefunction (11.4) satisfies (11.3) if and only if*

$$\frac{A_{21}}{A_{12}} = S(k_1 - k_2), \quad S(k) = \frac{k - ic}{k + ic}. \quad (11.5)$$

Proof. At $x_1 = x_2 = x$ both waves are $e^{i(k_1 + k_2)x}$, and $\partial_2 - \partial_1$ brings down $i(k_2 - k_1)$ from the first and $i(k_1 - k_2)$ from the second. With $k = k_1 - k_2$ the condition is $A_{12}(-ik - c) + A_{21}(ik - c) = 0$, so $A_{21}/A_{12} = (ik + c)/(ik - c) = (k - ic)/(k + ic)$. ■

S is [Proposition 10.1](#)'s formula at $\lambda = k/c$. No rapidity is needed: by Galilean invariance, which the lattice lacks, the phase depends on the momenta only through their difference. Its properties:

- $|S(k)| = 1$ for real k , and $S(k)S(-k) = 1$. Write

$$S(k) = -e^{i\theta(k)}, \quad \theta(k) = 2 \arctan \frac{k}{c}, \quad \theta'(k) = \frac{2c}{k^2 + c^2} > 0 \quad (11.6)$$

(Figure 11.2, left).

- $S(0) = -1$: with equal momenta $A_{21} = -A_{12}$ and the wave vanishes — a Pauli principle for bosons, which nobody put in.
- As $c \rightarrow \infty$ at fixed k , $S \rightarrow -1$: the wavefunction vanishes where two particles meet, and these *hard-core bosons* have the energies of free fermions (Girardeau 1960), the *Tonks–Girardeau* gas. On the ring the fermions obey $\psi(\dots, x_j+L, \dots) = (-1)^{N-1} \psi(\dots, x_j, \dots)$, periodic for odd N and antiperiodic for even N , and this is what “free fermions” means below. As $c \rightarrow 0$ at fixed $k \neq 0$, $S \rightarrow 1$: free bosons. (At $k = 0$, $S = -1$ for every c , so this limit is not uniform in k .)

Reading

Reflection and transmission. Were the two particles distinguishable, with the same interaction, the relative wave coming in from $r > 0$ would be partly transmitted to $r < 0$ and partly reflected, with amplitudes (the same integration)

$$t(k) = \frac{k}{k + ic}, \quad r(k) = \frac{-ic}{k + ic}, \quad |t|^2 + |r|^2 = 1.$$

For identical bosons a transmitted pair and a reflected pair are the same state, both amplitudes feed the one outgoing wave, and $t + r = S$.

What the phase does: a displacement

A plane wave has no position, so superpose: a narrow spread of k_1 around $\bar{k}_1$ and of k_2 around $\bar{k}_2 < \bar{k}_1$, the faster particle on the left. A packet sits where its phase is stationary in the momentum. In the incoming term the k_1 phase is $k_1 x_1 - k_1^2 t$, stationary at $x_1 = 2\bar{k}_1 t$. In the outgoing term it is $k_1 x_2 - k_1^2 t + \arg S(k_1 - k_2)$, and $\arg S = \pi + \theta$, so after the collision

• *sketched*

$$x_2 = 2\bar{k}_1 t - \theta'(\bar{k}_1 - \bar{k}_2), \quad x_1 = 2\bar{k}_2 t + \theta'(\bar{k}_1 - \bar{k}_2) \quad (11.7)$$

Each momentum comes out on a line parallel to its incoming one: the line carrying the larger momentum is shifted by $-\theta'$, the other by $+\theta'$ — in the centre-of-mass frame, both particles are delayed (Figure 11.2, right, checks this against numerically superposed exact solutions).

11.4 N bosons

The fundamental domain

Cut the ring at an origin, so that $0 \leq x_j < L$. The $\binom{N}{2}$ planes $x_i = x_j$ cut this cube into $N!$ sectors, one for each order of the particles; periodicity glues cyclically related sectors, which is where the Bethe equations will come from. A symmetric ψ is fixed by its values on the *fundamental domain* $x_1 < \dots < x_N$, whose images under permutations are the other sectors (Figure 11.3); it has $N - 1$ walls, $x_j = x_{j+1}$.

Definition 11.2 (Eigenfunctions of the delta gas). A symmetric, continuous function ψ , periodic with period L in every coordinate and smooth up to the walls within each sector, is an *eigenfunction* of (11.1) with eigenvalue E if $-\sum_j \partial_j^2 \psi = E\psi$ inside every sector, and on every wall $x_i = x_j$, approached from the side $x_i < x_j$, $(\partial_j - \partial_i - c)\psi = 0$.

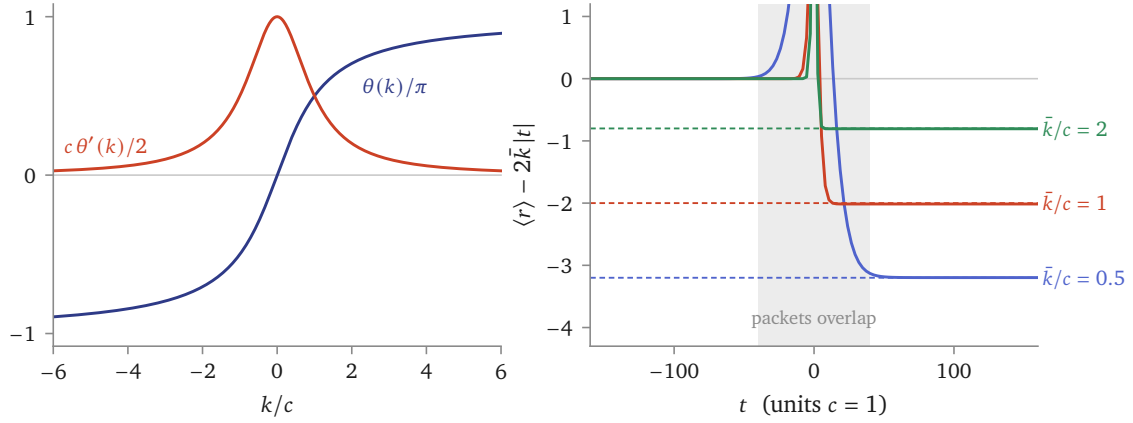


Figure 11.2: Left: the phase $\theta(k) = 2 \arctan(k/c)$, $S = -e^{i\theta}$, and its derivative. Right: two bosons, centre-of-mass frame, a Gaussian packet in the relative momentum, the exact two-body solution superposed numerically on $r = x_2 - x_1 > 0$. The centre $\langle r \rangle$ minus its hard-core value $2\bar{k}|t|$, $\bar{k} = \bar{k}_1 - \bar{k}_2$, starts at 0 and ends on $-2\langle\theta'(2q)\rangle$, averaged over the packet, close to the dashed narrow-packet value $-2\theta'(\bar{k})$ of (11.7); shaded, the packets overlap. (scripts/figures/lieb_liniger.py asserts both.)

The wall condition is § 11.3's, wall by wall. Rigorously H is defined as a self-adjoint operator through its quadratic form, in which the places where three particles meet have measure zero and impose nothing (the KLMN theorem; Reed and Simon, vol. II, §X.2) · cited.

The Bethe wavefunction

On the fundamental domain Bethe's ansatz is Equation (10.14) with momenta for rapidities:

$$\psi(x_1, \dots, x_N) = \sum_{\sigma \in S_N} A_\sigma \exp\left(i \sum_{j=1}^N k_{\sigma(j)} x_j\right), \quad A_\sigma = \prod_{\substack{a < b \\ \sigma \text{ puts } b \text{ before } a}} S(k_a - k_b), \quad (11.8)$$

one factor S for each pair of momenta that σ has exchanged.

Theorem 11.3 (Lieb and Liniger 1963). *Let $k_1, \dots, k_N$ be pairwise distinct, with no two differing by $\pm ic$, and let them satisfy the Bethe equations*

$$e^{ik_j L} = \prod_{l \neq j} \frac{k_j - k_l + ic}{k_j - k_l - ic}, \quad j = 1, \dots, N. \quad (11.9)$$

Then the symmetric extension of (11.8), if it is not zero, is an eigenfunction of the delta gas on the ring in the sense of Definition 11.2, with

$$E = \sum_{j=1}^N k_j^2, \quad K = \sum_{j=1}^N k_j. \quad (11.10)$$

Proof. Inside the sectors every term is a plane wave of energy $\sum k_j^2$. On the wall $x_{j+1} = x_j$ pair each σ with $\sigma' = \sigma \circ (j \ j+1)$, which exchanges k_a , $a = \sigma(j)$, and k_b , $b = \sigma(j+1)$. On the wall the two terms share an exponential, which no other term has, and together they are a

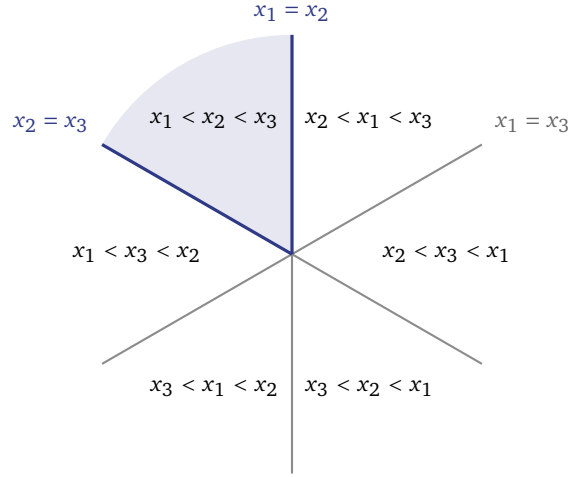


Figure 11.3: Three particles, seen along $(1, 1, 1)$: the planes $x_i = x_j$ cut the plane of relative coordinates into six sectors. The fundamental domain $x_1 < x_2 < x_3$ (shaded) has two walls; the other sectors are its images, and a symmetric wavefunction is fixed by its values there.

two-body wave in (x_j, x_{j+1}) times a factor free of x_j, x_{j+1} . By [Proposition 11.1](#) the contact condition holds if and only if $A_{\sigma'}/A_{\sigma} = S(k_a - k_b)$ for every σ ; the product in (11.8) gains or loses exactly the factor of the pair $\{a, b\}$, which is $S(k_a - k_b)$ either way since $S(k)S(-k) = 1$. The other walls are images of these, and the condition of [Definition 11.2](#) is symmetric, so it goes with them. Periodicity, $\psi(x_1, \dots, x_N) = \psi(x_2, \dots, x_N, x_1 + L)$, is the periodicity step of [Theorem 10.2](#) verbatim: moving k_a from last place to first passes it once past every other momentum, giving $e^{ik_a L} = \prod_{b \neq a} S(k_b - k_a)$, which is (11.9). ■

All roots are real, and each set of quantum numbers gives one state

The Heisenberg chain has complex roots, strings that make bound states (a reading of [Lecture 10](#)). Repulsion forbids them.

Proposition 11.4 (Real roots). *For $c > 0$, every solution of (11.9) is real.*

Proof. Let k_j have the largest imaginary part, m , and suppose $m > 0$. The left side of (11.9) has modulus $e^{-mL} < 1$. On the right, $\text{Im}(k_j - k_l) = d \geq 0$ for every l , and $|x + i(d + c)| \geq |x + i(d - c)|$ for real x because $d, c \geq 0$; so every factor has modulus at least 1, a contradiction. Hence $m \leq 0$. The same argument at the smallest imaginary part, with every inequality reversed, gives the smallest imaginary part ≥ 0 . So all are 0 (Korepin, Bogoliubov and Izergin, §I.2). ■

With real roots take logarithms, using $(k + ic)/(k - ic) = -e^{-i\theta(k)}$:

$$k_j L + \sum_{l=1}^N \theta(k_j - k_l) = 2\pi I_j, \quad I_j \in \mathbb{Z} + \frac{N-1}{2}, \quad (11.11)$$

the $N - 1$ signs going into I_j ($\theta(0) = 0$, so the term $l = j$ is harmless). Summing over j , the θ 's cancel in pairs and $K = (2\pi/L) \sum_j I_j$.

Proposition 11.5 (Yang and Yang 1969). For $c > 0$ and any real $I_1, \dots, I_N$, (11.11) has exactly one solution. If $I_j > I_l$ then $k_j > k_l$; in particular distinct quantum numbers give distinct roots, and equal quantum numbers give equal roots.

Proof (★). Let $\Theta(x) = \int_0^x \theta$, an even function growing at most linearly. Then (11.11) says $\partial B / \partial k_j = 0$ for

$$B(k) = \frac{L}{2} \sum_j k_j^2 - 2\pi \sum_j I_j k_j + \frac{1}{2} \sum_{j,l} \Theta(k_j - k_l),$$

whose Hessian is $L \mathbb{1} + \sum_{j < l} \theta'(k_j - k_l) (e_j - e_l)(e_j - e_l)^T$: L times the identity plus a sum of positive semidefinite matrices, since $\theta' > 0$. So B is strictly convex and grows like $\frac{L}{2} |k|^2$; it has exactly one critical point, its minimum. For the order, subtract the equations for j and l : $(k_j - k_l)L + \sum_m [\theta(k_j - k_m) - \theta(k_l - k_m)] = 2\pi(I_j - I_l)$. Since θ increases, the left side has the sign of $k_j - k_l$. ■

The reason: (11.11) is the gradient of a strictly convex function of $k_1, \dots, k_N$, which has exactly one minimum.

So for $c > 0$ the Bethe states are labelled exactly by sets of N distinct $I_j \in \mathbb{Z} + \frac{N-1}{2}$, one real solution each; coincident roots come only from coincident I 's, and then $S(0) = -1$ kills the vector. These are the quantum numbers of N free fermions, the $c \rightarrow \infty$ limit $k_j = 2\pi I_j / L$. That the Bethe states are *all* the eigenstates is here a theorem (Dorlas 1993, who also proves them orthogonal) · *cited*, unlike for the Heisenberg chain (a reading of [Lecture 10](#)). The ground state is the most compact set, $I_j = -\frac{N-1}{2}, \dots, \frac{N-1}{2}$, at every $c > 0$ (Yang and Yang 1969) · *cited*; [Figure 11.4](#) follows its roots.

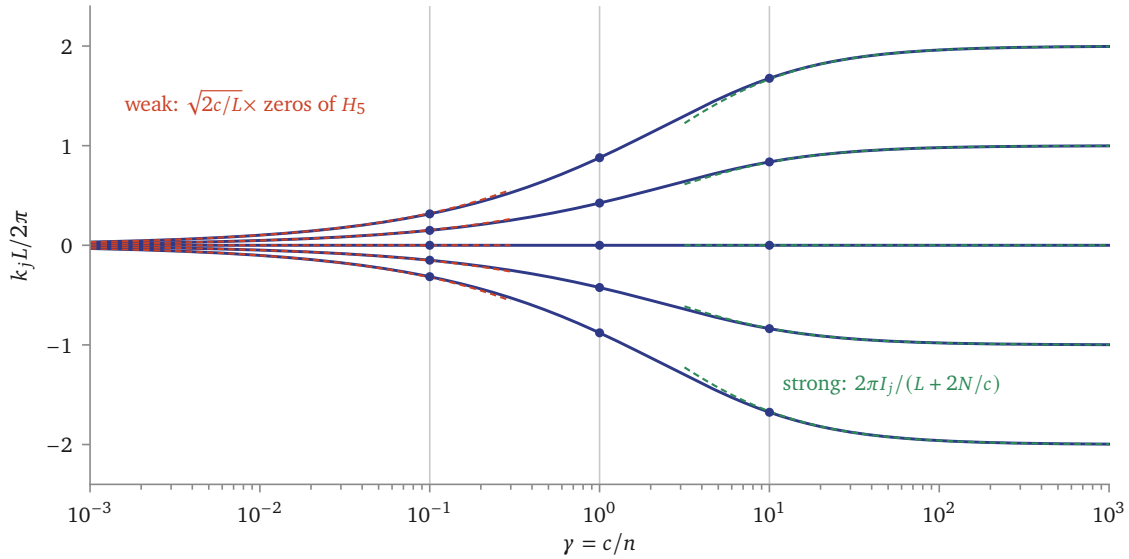


Figure 11.4: Ground-state roots of $N = 5$ bosons, $L = 5$ (density $n = 1$), against $\gamma = c/n$; dots at $\gamma = 0.1, 1, 10$. Strong coupling: towards the free-fermion $2\pi I_j / L$, as $2\pi I_j / (L + 2N/c)$ (green). Weak coupling: $\sqrt{2c/L}$ times the zeros of the Hermite polynomial H_5 (red); the tutorial derives both. (scripts/figures/lieb_liniger.py asserts the equations and both limits; random starts agree at the three marked couplings.)

Rapidities. On a ring the particles never become free, yet every eigenstate carries a set $\{k_j\}$, the *rapidities*, which enter E and K as if they were. The Bethe states form an orthogonal

basis (Dorlas 1993) · *cited*, so a state’s weight on each set $\{k_j\}$ is constant in time. That alone decides nothing about relaxation: the weights on the eigenstates of any Hamiltonian are constant. The distribution of § 0.2 is a coarser object. [Lecture 16](#) turns a large set $\{k_j\}$ into a density of rapidities. [Lecture 19](#) shows, for the Ising chain, how conserved mode occupations fix a non-thermal steady value of a local observable, and [Lecture 20](#) how a complete family of conserved charges fixes the rapidity density of a uniform gas. In the trap of the cradle that density is not conserved: [Lecture 20](#) follows its approximate motion, a shape that keeps returning.

11.5 Why one dimension

§ 0.2 counted. For two equal masses, momentum and energy are two equations for two outgoing momenta, solved only by the incoming pair or its exchange ([Proposition 0.1](#)). Quantum mechanically the same holds for *any* short-range pair interaction, once the particles are free again: the outgoing wave of two particles contains only the momenta k_1, k_2 , and scattering fixes nothing but amplitudes.

For three particles the count fails: two equations for three unknowns leave a circle of outgoing momenta, the six rearrangements being six points on it. A generic three-body collision sends waves out over the whole circle — *diffraction* — which is how a one-dimensional gas can thermalise (Sutherland, §2.2). **Non-diffraction is not free.** It is what the Bethe wavefunction asserts — in (11.8) only rearrangements of the incoming momenta ever appear — and [Theorem 11.3](#) proves that the delta gas has it. One caution: the two-body count uses $E = k^2$; on a lattice even two magnons are not forced by kinematics (§ 10.4, at $K = \pi$).

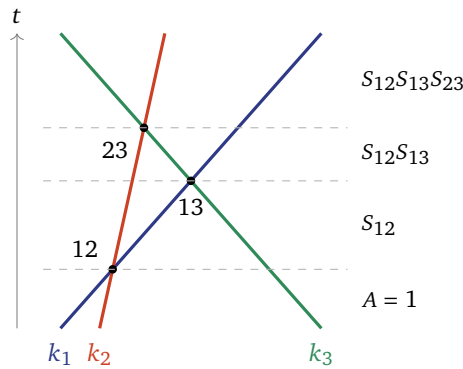


Figure 11.5: The Bethe wavefunction as a scattering event. Three momentum lines, $k_1 > k_2 > k_3$, cross pairwise; $S_{ab} = S(k_a - k_b)$. For a superposition of separated wave packets, between collisions the packet is carried by one term of (11.8), with amplitude A_σ (right): one factor per crossing passed. No other momenta appear. [Lecture 12](#) draws the same picture for the Toda chain, with a displacement where here there is a phase; the other order of crossings gives the same product, because numbers commute.

In [Figure 11.5](#) each crossing multiplies the amplitude by one S . Had the lines crossed in the other order, the same factors would come in another order, and nothing would change: for bosons without labels three-body consistency is automatic. It stops being automatic when a collision can do more than multiply by a number, which is [Lecture 13](#)’s subject.

11.6 What it buys

- *An interacting gas solved exactly*, with a complete proof, and every state labelled by N distinct quantum numbers, as for free fermions.
- *A clean thermodynamic limit*. Real roots and one state per set of quantum numbers are what [Lecture 16](#) and [Lecture 17](#) need: the roots become a density, with no strings to track.
- *Rapidities that label every state*, which [Lecture 16](#) turns into a density: conserved in a uniform gas, where [Lecture 20](#) shows what it fixes ([Lecture 19](#) does the same for the Ising chain's occupations), and moving in the cradle's trap, where [Lecture 20](#) follows it.

11.7 Caveats, in one place

- *Status*. [Theorem 11.3](#) is proved, wall by wall: unlike on the lattice, where three adjacent spins are a real configuration, the walls of [Definition 11.2](#) never meet in the conditions. [Proposition 11.4](#) and [Proposition 11.5](#) are proved. Completeness and orthogonality of the Bethe states (Dorlas 1993), the ground state's quantum numbers (Yang and Yang 1969) and the delta as a model of the tubes (Olshanii 1998) are cited.
- *Repulsion is needed*. For attractive $c < 0$ the argument of [Proposition 11.4](#) fails: complex roots appear, in strings, as bound states of bosons (McGuire 1964). That gas is not treated.

Tutorial

11.8 Solving the equations

Two particles by hand. Put $k_{1,2} = K/2 \pm q$ with $K = 2\pi n/L$. Then $e^{iKL/2} = (-1)^n$, and the first equation of (11.9) becomes $qL + \theta(2q) = \pi(n+1) \pmod{2\pi}$, one equation for $q > 0$ ($q = 0$ gives the zero vector). Its left side increases from 0, so each allowed right side gives one root, and $E = K^2/2 + 2q^2$. For $K = 0$ the ground state has $qL + 2 \arctan(2q/c) = \pi$: $q \rightarrow \pi/L$ as $c \rightarrow \infty$ (free fermions, $k = \pm\pi/L$) and $q \approx \sqrt{c/L}$ as $c \rightarrow 0$. The check script diagonalises two bosons on a ring of M sites, $-\Delta/a^2 + (2c/a)\delta$, with $a = L/M$, at $M = 200, 400, 800$, extrapolates in a , and finds the twelve lowest levels to within 6×10^{-9} of these.

Numerically, use damped Newton on (11.11): its Jacobian is the Hessian of the convex function B of [Proposition 11.5](#), positive definite, so every step is well defined; halve the step until the residual decreases. Start from the free-fermion values $2\pi I_j/L$.

Strong coupling. For $|k| \ll c$, $\theta(k) \approx 2k/c$, and with $\sum_l k_l = 0$ for the symmetric ground state (11.11) becomes $k_j(L + 2N/c) = 2\pi I_j$, with error $O(c^{-3})$: the gas behaves as free fermions on a ring lengthened by $2N/c$.

Weak coupling. For $cL \ll 1$ the roots of the ground state crowd together, but their differences, of order $\sqrt{c/L}$, stay large compared with c ; then $\theta(x) \approx \pi \operatorname{sgn} x - 2c/x$. The $\pi \operatorname{sgn}$ terms add up to exactly $2\pi I_j$ for the symmetric I 's and cancel, leaving $k_j L = \sum_{l \neq j} 2c/(k_j - k_l)$. With $k_j = \sqrt{2c/L} h_j$ this is $h_j = \sum_{l \neq j} 1/(h_j - h_l)$, Stieltjes' characterisation of the zeros of the Hermite polynomial H_N (Gaudin §4.3). [Figure 11.4](#) shows both limits.

Summary.

- *The gas*: N bosons on a ring with a contact interaction (11.1), the gas of the quantum Newton’s cradle.
- *Two bosons*: the contact condition gives the phase $S = -e^{i\theta}$, $\theta(k) = 2 \arctan(k/c)$ (Proposition 11.1).
- *N bosons*: the Bethe equations (11.9), with a complete proof (Theorem 11.3); for $c > 0$ every root is real (Proposition 11.4), and each set of quantum numbers has exactly one solution (Proposition 11.5).
- *Why one dimension*: two-body collisions keep the set of momenta, and the Bethe wavefunction shows that three-body collisions do not diffract.
- *Next*: Lecture 12 watches the same non-diffraction in classical scattering, on the open Toda chain.

Tutorial. Exercise Sheet 11.

References

E. H. Lieb and W. Liniger, Phys. Rev. 130 (1963) 1605; C. N. Yang and C. P. Yang, J. Math. Phys. 10 (1969) 1115; M. Olshanii, Phys. Rev. Lett. 81 (1998) 938; T. C. Dorlas, Commun. Math. Phys. 154 (1993) 347; Korepin, Bogoliubov and Izergin, *Quantum Inverse Scattering Method and Correlation Functions*, §I.2; Sutherland, *Beautiful Models*, §§2.2, 4.2; Gaudin, *The Bethe Wavefunction*, §4.3.